



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR Symbiosis International (Deemed University)

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Practical No. : 2

Aim -

Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Multi-Item E-Receipt</title>
<style>
  body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    margin: 0;
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background: #f4f7f6;
  }
  .container {
    max-width: 800px;
    width: 95%;
    background: #fff;
    padding: 30px;
    border-radius: 10px;
    box-shadow: 0 6px 20px rgba(0,0,0,0.3);
  }
  h2 {
    text-align: center;
    color: #2e7d32; /* Green heading */
    margin-bottom: 20px;
  }
  label {
```

```

display: block;
margin-top: 12px;
font-weight: bold;
color: #333;
}
input {
width: 100%;
padding: 10px;
margin-top: 5px;
border: 1px solid #aaa;
border-radius: 6px;
}
input:focus {
border-color: #2e7d32;
outline: none;
}
button {
margin-top: 15px;
padding: 12px 20px;
background: #2e7d32; /* Green button */
color: #fff;
border: none;
font-size: 14px;
border-radius: 6px;
cursor: pointer;
transition: 0.3s ease;
}
button:hover {
background: #1b5e20;
}
table {
width: 100%;
border-collapse: collapse;
margin-top: 15px;
}
table, th, td {
border: 1px solid #2e7d32;
}
th, td {
padding: 10px;
text-align: center;
color: #1b5e20; /* Green font */
}
th {
background: #e8f5e9;
}
#receipt {
margin-top: 25px;
padding: 20px;
border: 2px solid #2e7d32;
border-radius: 8px;
background: #e8f5e9;
color: #1b5e20;
}
#receipt h3 {
text-align: right;
font-size: 18px;
margin-top: 15px;
color: #2e7d32;
}
</style>
</head>
<body>

<div class="container">

```

```
<h2>Multi-Item E-Receipt</h2>
```

```
<label>Customer Name:</label>
```

```
<input type="text" id="customerName" placeholder="Enter customer name">
```

```
<h3>Add Product</h3>
```

```
<label>Product Name:</label>
```

```
<input type="text" id="productName" placeholder="Enter product name">
```

```
<label>Price (₹):</label>
```

```
<input type="number" id="price" placeholder="Enter price">
```

```
<label>Quantity:</label>
```

```
<input type="number" id="quantity" placeholder="Enter quantity">
```

```
<button onclick="addItem()">Add Item</button>
```

```
<button onclick="generateReceipt()">Generate Receipt</button>
```

```
<div id="cart"></div>
```

```
<div id="receipt"></div>
```

```
</div>
```

```
<script>
```

```
let items = [];
```

```
function addItem() {
```

```
  const productName = document.getElementById("productName").value;
```

```
  const price = parseFloat(document.getElementById("price").value);
```

```
  const quantity = parseInt(document.getElementById("quantity").value);
```

```
  if (!productName || isNaN(price) || price <= 0 || isNaN(quantity) || quantity <= 0) {  
    alert("Please enter valid product details.");
```

```
    return;
```

```
  }
```

```
  // GST per item
```

```
  const subtotal = price * quantity;
```

```
  const gstAmount = subtotal * 0.018;
```

```
  const total = subtotal + gstAmount;
```

```
  items.push({ productName, price, quantity, subtotal, gstAmount, total });
```

```
  displayCart();
```

```
  // Clear inputs
```

```
  document.getElementById("productName").value = "";
```

```
  document.getElementById("price").value = "";
```

```
  document.getElementById("quantity").value = "";
```

```
}
```

```
function displayCart() {
```

```
  let cartHTML = "<h3>Items
```

```
Added</h3><table><tr><th>Name</th><th>Price</th><th>Qty</th><th>Subtotal</th><th>GST  
(18%)</th><th>Total</th></tr>";
```

```
  items.forEach(item => {
```

```
    cartHTML += `<tr>
```

```
      <td>${item.productName}</td>
```

```
      <td>₹${item.price.toFixed(2)}</td>
```

```
      <td>${item.quantity}</td>
```

```
      <td>₹${item.subtotal.toFixed(2)}</td>
```

```
      <td>₹${item.gstAmount.toFixed(2)}</td>
```

```
      <td>₹${item.total.toFixed(2)}</td>
```

```
    </tr>`;
```

```
  });
```

```
  cartHTML += "</table>";
```

```

document.getElementById("cart").innerHTML = cartHTML;
}

function generateReceipt() {
  const customerName = document.getElementById("customerName").value;
  if (!customerName || items.length === 0) {
    alert("Please enter customer name and add at least one item.");
    return;
  }

  const grandSubtotal = items.reduce((sum, item) => sum + item.subtotal, 0);
  const grandGST = items.reduce((sum, item) => sum + item.gstAmount, 0);
  const grandTotal = items.reduce((sum, item) => sum + item.total, 0);

  document.getElementById("receipt").innerHTML = `
    <h2>E-Receipt</h2>
    <p><b>Customer:</b> ${customerName}</p>
    <p><b>Total Subtotal:</b> ₹${grandSubtotal.toFixed(2)}</p>
    <p><b>Total GST (18%):</b> ₹${grandGST.toFixed(2)}</p>
    <h3>Grand Total: ₹${grandTotal.toFixed(2)}</h3>
  `;
}
</script>

</body>
</html>

```

Output -

Multi-Item E-Receipt

Customer Name:

sejal

Add Product

Product Name:

Enter product name

Price (₹):

Enter price

Quantity:

Enter quantity

Add Item

Generate Receipt

Items Added

Name	Price	Qty	Subtotal	GST (18%)	Total
soap	₹55.00	4	₹220.00	₹3.96	₹223.96
towel	₹775.00	8	₹6200.00	₹111.60	₹6311.60

E-Receipt

Customer: sejal

Total Subtotal: ₹6420.00

Total GST (18%): ₹115.56

Grand Total: ₹6535.56

Case Study Title –

Develop the webpage for an online shopping website with using HTML, CSS and HTML tags and calculate the bill.

Case Study Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Shopping Paradise</title>
  <style>
    body {
```

```

    font-family: Arial, sans-serif;
    margin: 0;
    background-color: #f5f5f5;
}
header {
    background: linear-gradient(90deg, #0b1d3a, #1f4a7a, #6c7a89, #f5e9d7);
    color: white;
    padding: 25px;
    text-align: center;
    font-size: 32px;
    font-weight: bold;
    letter-spacing: 2px;
    text-transform: uppercase;
}
.container {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    margin: 20px;
}
.product {
    border: 1px solid #ddd;
    background: #f5e9d7;
    margin: 15px;
    padding: 15px;
    width: 250px;
    text-align: center;
    box-shadow: 0 2px 8px rgba(0,0,0,0.2);
    border-radius: 8px;
}
.product h3 {
    color: #1f4a7a;
}
.product button {
    background: #0b1d3a;
    border: none;
    padding: 10px 15px;
    margin-top: 10px;
    cursor: pointer;
    color: #fff;
    font-weight: bold;
    border-radius: 4px;
}
.product button:hover {
    background: #1f4a7a;
}
#bill {
    background: #fff;
    margin: 30px auto;
    padding: 20px;
    width: 600px;
    border: 2px solid #4CAF50;
    box-shadow: 0 3px 8px rgba(0,0,0,0.2);
    border-radius: 8px;
}
#bill h2 {
    margin-top: 0;
    text-align: center;
    color: #0b1d3a;
    font-size: 24px;
}
#bill table {
    width: 100%;
    border-collapse: collapse;
}

```

```

#bill table, #bill th, #bill td {
    border: 1px solid #ccc;
}
#bill th, #bill td {
    padding: 10px;
    text-align: center;
}
#total {
    font-weight: bold;
    color: #4CAF50;
}
#gst {
    font-weight: bold;
    color: #d9534f;
}
#grandTotal {
    font-weight: bold;
    color: #1f4a7a;
    font-size: 20px;
}
.footer {
    margin-top: 20px;
    text-align: center;
    font-size: 20px;
    font-weight: bold;
    border-top: 2px dashed #333;
    padding-top: 10px;
}
</style>
</head>
<body>
<header>
    Shopping Paradise
</header>

<!-- Customer Name Input -->
<div style="text-align:center; margin:20px;">
    <label>Enter Customer Name:
        <input type="text" id="customerName" placeholder="Customer Name">
    </label>
</div>

<div class="container">
    <!-- Products -->
    <div class="product">
        <h3>Shirt</h3>
        <p>Price: ₹500</p>
        <label>Qty: <input type="number" id="qtyShirt" value="1" min="1"></label><br>
        <button onclick="addToBill('Shirt', 500, document.getElementById('qtyShirt').value)">Add</button>
    </div>

    <div class="product">
        <h3>Jeans</h3>
        <p>Price: ₹1200</p>
        <label>Qty: <input type="number" id="qtyJeans" value="1" min="1"></label><br>
        <button onclick="addToBill('Jeans', 1200, document.getElementById('qtyJeans').value)">Add</button>
    </div>

    <div class="product">
        <h3>Necklace</h3>
        <p>Price: ₹3000</p>
        <label>Qty: <input type="number" id="qtyNecklace" value="1" min="1"></label><br>
        <button onclick="addToBill('Necklace', 3000,
document.getElementById('qtyNecklace').value)">Add</button>
    </div>

```

```

<div class="product">
  <h3>Ring</h3>
  <p>Price: ₹1500</p>
  <label>Qty: <input type="number" id="qtyRing" value="1" min="1"></label><br>
  <button onclick="addToBill('Ring', 1500, document.getElementById('qtyRing').value)">Add</button>
</div>

<div class="product">
  <h3>Mobile</h3>
  <p>Price: ₹15000</p>
  <label>Qty: <input type="number" id="qtyMobile" value="1" min="1"></label><br>
  <button onclick="addToBill('Mobile', 15000,
document.getElementById('qtyMobile').value)">Add</button>
</div>

<div class="product">
  <h3>Laptop</h3>
  <p>Price: ₹45000</p>
  <label>Qty: <input type="number" id="qtyLaptop" value="1" min="1"></label><br>
  <button onclick="addToBill('Laptop', 45000,
document.getElementById('qtyLaptop').value)">Add</button>
</div>
</div>

<!-- Bill Section -->
<div id="bill">
  <h2>Bill Details</h2>
  <p id="customerLine" style="text-align:center; font-weight:bold;"></p>
  <table id="billTable">
    <tr>
      <th>Item</th>
      <th>Qty</th>
      <th>Price</th>
      <th>Subtotal</th>
    </tr>
  </table>
  <p id="total">Total: ₹0</p>
  <p id="gst">GST (1.8%): ₹0</p>
  <p id="grandTotal">Grand Total: ₹0</p>
  <div class="footer" id="footerMessage">
    <!-- Thank you message will appear here -->
  </div>
</div>

<script>
  let total = 0;

  function addToBill(item, price, qty) {
    qty = parseInt(qty);
    let subtotal = price * qty;

    // Add row to bill table
    let table = document.getElementById("billTable");
    let row = table.insertRow();
    row.insertCell(0).textContent = item;
    row.insertCell(1).textContent = qty;
    row.insertCell(2).textContent = "₹" + price;
    row.insertCell(3).textContent = "₹" + subtotal;

    // Update totals
    total += subtotal;
    let gst = total * 0.018;
    let grandTotal = total + gst;

```

```

document.getElementById("total").textContent = "Total: ₹" + total.toFixed(2);
document.getElementById("gst").textContent = "GST (1.8%): ₹" + gst.toFixed(2);
document.getElementById("grandTotal").textContent = "Grand Total: ₹" + grandTotal.toFixed(2);

// Show customer name on bill
let name = document.getElementById("customerName").value;
if(name.trim() !== "") {
    document.getElementById("customerLine").textContent = "Customer Name: " + name;
    document.getElementById("footerMessage").innerHTML = "<p><b>Thank you " + name + " for shopping with us.</b></p><p>Visit Again</p>";
}
}
</script>
</body>
</html>

```

Output –

SHOPPING PARADISE

Enter Customer Name

Shirt

Price: ₹500

Qty:

Add

Jeans

Price: ₹1200

Qty:

Add

Necklace

Price: ₹3000

Qty:

Add

Ring

Price: ₹1500

Qty:

Add

Mobile

Price: ₹15000

Qty:

Add

Laptop

Price: ₹45000

Qty:

Add

Bill Details

Customer Name: sejal lende

Item	Qty	Price	Subtotal
Shirt	1	₹500	₹500
Ring	1	₹1500	₹1500
Mobile	1	₹15000	₹15000
Mobile	1	₹15000	₹15000
Mobile	1	₹15000	₹15000
Mobile	1	₹15000	₹15000

Total: ₹62000.00

GST (1.8%): ₹1116.00

Grand Total: ₹63116.00

Thank you sejal lende for shopping with us.

Visit Again

Conclusion –

The experiment was successfully performed to understand JavaScript variables, data types, template literals, and destructuring. These concepts were applied to develop an online shopping billing calculator that accepts user input and generates the final invoice dynamically.